

Assignment 5

Week 5

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Problem 1. Finding Unitary Matrix by Hand

Construct the matrix for the transformation from the basis of σ_y to σ_x . In other words, finding the unitary matrix to re-represent a vector in the σ_y eigenvector basis representation in the σ_x eigenvector basis representation.

Solution

The two eigenbases. The normalized eigenvectors, with $\sigma_x |x_{\pm}\rangle = \pm |x_{\pm}\rangle$ and $\sigma_y |y_{\pm}\rangle = \pm |y_{\pm}\rangle$, are

$$|x_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |x_-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad |y_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |y_-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

What the matrix must do. If $|\psi\rangle$ has coordinates $a_j = \langle y_j | \psi \rangle$ in the σ_y eigenbasis and $b_i = \langle x_i | \psi \rangle$ in the σ_x eigenbasis, then inserting $\sum_j |y_j\rangle \langle y_j| = I$ gives

$$b_i = \langle x_i | \psi \rangle = \sum_j \langle x_i | y_j \rangle \langle y_j | \psi \rangle = \sum_j U_{ij} a_j, \quad U_{ij} = \langle x_i | y_j \rangle.$$

Entries (each $|x_i\rangle$ is real, so $\langle x_i | y_j \rangle = \frac{1}{2} x_i^\top y_j$):

$$\begin{aligned} U_{++} &= \frac{1}{2}(1+i), & U_{+-} &= \frac{1}{2}(1-i), \\ U_{-+} &= \frac{1}{2}(1-i), & U_{--} &= \frac{1}{2}(1+i), \end{aligned}$$

so

$$U = \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}.$$

Unitarity check.

$$U^\dagger U = \frac{1}{4} \begin{pmatrix} 1-i & 1+i \\ 1+i & 1-i \end{pmatrix} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = I,$$

so U is unitary, as any change between orthonormal bases must be.

Problem 2. Completeness of Basis

Show that the eigenvectors of σ_x form a complete basis.

(*Hint:* see the following equation.)

Solution

The eigenvectors of σ_x ,

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

are orthonormal: $\langle + | + \rangle = \langle - | - \rangle = 1$ and $\langle + | - \rangle = \frac{1}{2}(1-1) = 0$. An orthonormal set is a complete basis of $\mathbb{C}^2$ exactly when it satisfies the completeness relation $\sum_i |i\rangle \langle i| = I$.

$$\begin{aligned} |+\rangle\langle+| &= \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} |-\rangle\langle-| &= \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \end{aligned}$$

Adding the two projectors,

$$|+\rangle\langle+| + |-\rangle\langle-| = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I.$$

The identity therefore resolves over $\{|+\rangle, |-\rangle\}$, so every $|\psi\rangle \in \mathbb{C}^2$ expands as $|\psi\rangle = I|\psi\rangle = \langle+|\psi\rangle|+\rangle + \langle-|\psi\rangle|-\rangle$: the eigenvectors of σ_x form a complete basis. ■

Problem 3. Construction of Matrix

Construct the σ_x by using its eigenvalues and eigenvectors.

Solution

The spectral decomposition rebuilds a matrix from its eigenvalues λ_i and eigen-projectors: $\sigma_x = \sum_i \lambda_i |x_i\rangle\langle x_i|$. With eigenpairs $(+1, |+\rangle)$ and $(-1, |-\rangle)$ and the projectors from **Problem 2**,

$$\begin{aligned} \sigma_x &= (+1) |+\rangle\langle+| + (-1) |-\rangle\langle-| \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \end{aligned}$$

which is exactly σ_x . ■

Problem 4. Colab Verification

Use Colab to verify the results from Problems 1 to 3.

Solution

We reproduce Problems 1–3 in Colab: assemble $U_{ij} = \langle x_i | y_j \rangle$ with `np.vdot` and test $U^\dagger U = I$; form $\sum_i |x_i\rangle\langle x_i|$ and the spectral sum $\sum_i \lambda_i |x_i\rangle\langle x_i|$ with `np.outer`.

```
1 xs = [np.array([1, 1]) / np.sqrt(2), np.array([1, -1]) / np.sqrt(2)] # |x+>, |x->
2 ys = [np.array([1, 1j]) / np.sqrt(2), np.array([1, -1j]) / np.sqrt(2)] # |y+>, |y->
3
4 U = np.array([[np.vdot(xi, yj) for yj in ys] for xi in xs]) # U_ij = <x_i|y_j>
5 print(np.allclose(U.conj().T @ U, np.eye(2))) # Problem 1
6
7 completeness = sum(np.outer(xi, xi.conj()) for xi in xs) # Problem 2
8 spectral = sum(l * np.outer(xi, xi.conj()) for l, xi in zip((1, -1), xs)) # Problem 3
9 print(np.allclose(completeness, np.eye(2)),
10       np.allclose(spectral, [[0, 1], [1, 0]]))
```

The notebook (**Appendix 1**) prints True for every check: $U = \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$ is unitary, $|+\rangle\langle+| + |-\rangle\langle-| = I$, and $(+1)|+\rangle\langle+| + (-1)|-\rangle\langle-| = \sigma_x$ – confirming Problems 1, 2, and 3.

Problem 5. Computing Tensor Product

Let $|f\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $|g\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix}$, find $|f\rangle_1 \otimes |g\rangle_2$ and $|g\rangle_1 \otimes |f\rangle_2$. Are they the same? Here you see the order is very important.

Solution

Writing $|f\rangle = (0, 1)^\top$ and $|g\rangle = \left(\frac{1}{\sqrt{2}}, \frac{i}{\sqrt{2}}\right)^\top$, the tensor product places one scaled copy of the second factor under each entry of the first:

$$|f\rangle_1 \otimes |g\rangle_2 = \begin{pmatrix} 0 \cdot |g\rangle \\ 1 \cdot |g\rangle \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix} \quad \left| \quad |g\rangle_1 \otimes |f\rangle_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \cdot |f\rangle \\ \frac{i}{\sqrt{2}} \cdot |f\rangle \end{pmatrix} = \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 0 \\ i/\sqrt{2} \end{pmatrix}$$

They are **not** the same (compare the second entries, 0 against $1/\sqrt{2}$). The tensor product does not commute: the order of the factors records which subsystem is which, so $|f\rangle \otimes |g\rangle$ and $|g\rangle \otimes |f\rangle$ are genuinely different states.

Problem 6. Computing Tensor Products of Three Vectors

Find $\begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. What is the dimension of this space?

Solution

Associating left to right, the first two factors give $\left(0, 0, \frac{1}{\sqrt{2}}, \frac{i}{\sqrt{2}}\right)^\top$ (as in **Problem 5**). Tensoring that 4-vector with $(0, 1)^\top$ sends each entry $c \mapsto (0, c)^\top$:

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1/\sqrt{2} \\ 0 \\ i/\sqrt{2} \end{pmatrix}^\top.$$

The space is $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$, of dimension $2 \times 2 \times 2 = 8$.

Problem 7. Distribution Law

Check the following equation:

$$(|f\rangle + |e\rangle) \otimes |g\rangle = |f\rangle \otimes |g\rangle + |e\rangle \otimes |g\rangle$$

$$|g\rangle \otimes (|f\rangle + |e\rangle) = |g\rangle \otimes |f\rangle + |g\rangle \otimes |e\rangle$$

by substituting $|f\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $|e\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and $|g\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix}$ to both sides.

Solution

Let $|f\rangle = (0, 1)^\top$, $|e\rangle = (1, 0)^\top$, $|g\rangle = (s, is)^\top$ with $s = \frac{1}{\sqrt{2}}$, so $|f\rangle + |e\rangle = (1, 1)^\top$.

Left distributivity.

$$\text{LHS: } (|f\rangle + |e\rangle) \otimes |g\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} s \\ is \end{pmatrix} = (s, is, s, is)^\top,$$

$$\text{RHS: } |f\rangle \otimes |g\rangle + |e\rangle \otimes |g\rangle = (0, 0, s, is)^\top + (s, is, 0, 0)^\top = (s, is, s, is)^\top.$$

Right distributivity.

$$\text{LHS: } |g\rangle \otimes (|f\rangle + |e\rangle) = \begin{pmatrix} s \\ is \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (s, s, is, is)^\top,$$

$$\text{RHS: } |g\rangle \otimes |f\rangle + |g\rangle \otimes |e\rangle = (0, s, 0, is)^\top + (s, 0, is, 0)^\top = (s, s, is, is)^\top.$$

In each case LHS = RHS, so the tensor product distributes over vector addition from both sides.

Problem 8. Tensor Products using Colab

Use Colab to verify the results from Problems 5 to 7. For Problem 5, you may use this code

```
1 import numpy as np
2
3 g = np.array([[0], [1]])
4 h = np.array([[1/np.sqrt(2)], [1j/np.sqrt(2)]])
5
6 print(np.kron(g, h))
7 print(np.kron(h, g))
```

Solution

Checking Problems 5–7 with np.kron:

```
1 f = np.array([0, 1]); e = np.array([1, 0]); g = np.array([1, 1j]) / np.sqrt(2)
2
3 print(np.kron(f, g), np.kron(g, f)) # Problem 5
4 print(np.kron(np.kron(f, g), f).shape) # Problem 6 -> (8,)
5 print(np.allclose(np.kron(f + e, g), np.kron(f, g) + np.kron(e, g))) # Problem 7 (left)
6 print(np.allclose(np.kron(g, f + e), np.kron(g, f) + np.kron(g, e))) # Problem 7 (right)
```

The notebook (**Appendix 1**) shows, with $s = \frac{1}{\sqrt{2}}$, $|f\rangle \otimes |g\rangle = (0, 0, s, is)^\top \neq (0, s, 0, is)^\top = |g\rangle \otimes |f\rangle$ (Problem 5), a three-fold product of shape (8,) (Problem 6), and True for both distributive laws (Problem 7).

Problem 9. Inner Product of Tensor Products

Check the following equation:

$$\langle h_1 | h_2 \rangle = \left\langle \sum_{j=1}^{n_1} |f_j\rangle \otimes |g_j\rangle \left| \sum_{l=1}^{n_2} |e_l\rangle \otimes |k_l\rangle \right. \right\rangle = \sum_{j=1}^{n_1} \sum_{l=1}^{n_2} \langle f_j | e_l \rangle \langle g_j | k_l \rangle$$

by setting $n_1 = n_2 = 1$ (i.e. only one term) and $|f\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $|e\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $|g\rangle = \begin{pmatrix} i \\ i \end{pmatrix}$, and $|k\rangle = \begin{pmatrix} i \\ 1 \end{pmatrix}$.

Solution

Why it holds. For simple tensors the inner product on $H_1 \otimes H_2$ factorizes,

$$\langle a \otimes b | c \otimes d \rangle = (a \otimes b)^\dagger (c \otimes d) = (a^\dagger c) (b^\dagger d) = \langle a | c \rangle \langle b | d \rangle.$$

Expanding both sums by (conjugate-)linearity,

$$\left\langle \sum_{j=1}^{n_1} |f_j\rangle \otimes |g_j\rangle \left| \sum_{l=1}^{n_2} |e_l\rangle \otimes |k_l\rangle \right. \right\rangle = \sum_{j=1}^{n_1} \sum_{l=1}^{n_2} \langle f_j \otimes g_j | e_l \otimes k_l \rangle = \sum_{j=1}^{n_1} \sum_{l=1}^{n_2} \langle f_j | e_l \rangle \langle g_j | k_l \rangle.$$

Check with $n_1 = n_2 = 1$. Here $|h_1\rangle = |f\rangle \otimes |g\rangle$ and $|h_2\rangle = |e\rangle \otimes |k\rangle$, with $|f\rangle = (1, 1)^\top$, $|e\rangle = (0, 1)^\top$, $|g\rangle = (i, i)^\top$, $|k\rangle = (i, 1)^\top$, so

$$|h_1\rangle = (i, i, i, i)^\top, \quad |h_2\rangle = (0, 0, i, 1)^\top.$$

$$\begin{aligned}\langle h_1|h_2\rangle &= \begin{pmatrix} -i & -i & -i & -i \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ i \\ 1 \end{pmatrix} \\ &= (-i)(i) + (-i)(1) \\ &= 1 - i\end{aligned}$$

$$\begin{aligned}\langle f|e\rangle &= (1)(0) + (1)(1) = 1, \\ \langle g|k\rangle &= (-i)(i) + (-i)(1) = 1 - i, \\ \langle f|e\rangle \langle g|k\rangle &= (1)(1 - i) = 1 - i\end{aligned}$$

Both routes give $\langle h_1|h_2\rangle = 1 - i$, so the equation checks out. ■

Problem 10. Hand Calculation of Tensor Product

Let $|f\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $|g\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix}$, find $\sigma_x |f\rangle_1$ and $\sigma_x |g\rangle_2$. Then find the tensor product of the resulting vector (i.e. $\sigma_x |f\rangle_1 \otimes \sigma_x |g\rangle_2$). Now find $|f\rangle_1 \otimes |g\rangle_2$ and construct $\sigma_x \otimes \sigma_x$. Show that $\sigma_x \otimes \sigma_x |f\rangle_1 \otimes \sigma_x |g\rangle_2$ gives the same result.

Solution

Let $s = \frac{1}{\sqrt{2}}$. σ_x swaps a vector's two entries, so

$$\sigma_x |f\rangle_1 = \sigma_x \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \sigma_x |g\rangle_2 = \sigma_x \begin{pmatrix} s \\ is \end{pmatrix} = \begin{pmatrix} is \\ s \end{pmatrix}.$$

Transform each factor, then tensor:

$$\sigma_x |f\rangle_1 \otimes \sigma_x |g\rangle_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} is \\ s \end{pmatrix} = (is, s, 0, 0)^\top.$$

Tensor first, then transform: from **Problem 5**, $|f\rangle_1 \otimes |g\rangle_2 = (0, 0, s, is)^\top$, and

$$\sigma_x \otimes \sigma_x = \begin{pmatrix} 0\sigma_x & 1\sigma_x \\ 1\sigma_x & 0\sigma_x \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

which reverses the entry order of a 4-vector, so

$$(\sigma_x \otimes \sigma_x) (|f\rangle_1 \otimes |g\rangle_2) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ s \\ is \end{pmatrix} = (is, s, 0, 0)^\top.$$

Both give $(is, s, 0, 0)^\top$ – the mixed-product property $(\sigma_x \otimes \sigma_x) (|f\rangle \otimes |g\rangle) = (\sigma_x |f\rangle) \otimes (\sigma_x |g\rangle)$.

Problem 11. Tensor Product Calculation using Colab

Use Colab and the function `np.kron` to verify Problem 10.

Solution

Verifying Problem 10 with `np.kron` for both the states and the operator:

```
1 sx = np.array([[0, 1], [1, 0]])
2 f = np.array([0, 1]); g = np.array([1, 1j]) / np.sqrt(2)
3
4 lhs = np.kron(sx @ f, sx @ g)          # (sx|f>) (x) (sx|g>)
5 rhs = np.kron(sx, sx) @ np.kron(f, g) # (sx (x) sx)(|f> (x) |g>)
6 print(lhs)
7 print(np.allclose(lhs, rhs))
```

The notebook (**Appendix 1**) gives $\text{lhs} = (is, s, 0, 0)^\top$ with $s = \frac{1}{\sqrt{2}}$ and $\text{np.allclose}(\text{lhs}, \text{rhs}) = \text{True}$: acting with $\sigma_x \otimes \sigma_x$ on $|f\rangle \otimes |g\rangle$ reproduces $(\sigma_x |f\rangle) \otimes (\sigma_x |g\rangle)$, confirming Problem 10.

APPENDIX 1

Assignment 05 | Code Solution

assignment-05-code

September 9, 2026

1 Assignment 05

1.1 Import External Libraries

```
[1]: import numpy as np
import sympy as sp

from IPython.display import display, Math
```

1.2 4. Colab Verification (Problems 1–3)

```
[2]: # --- Problem 4: verify Problems 1-3 in Colab ---
xs = [np.array([1, 1]) / np.sqrt(2), np.array([1, -1]) / np.sqrt(2)] # |x+>, |x->
ys = [np.array([1, 1j]) / np.sqrt(2), np.array([1, -1j]) / np.sqrt(2)] # |y+>, |y->

# Problem 1: change of basis U_ij = <x_i | y_j>
U = np.array([np.vdot(xi, yj) for yj in ys for xi in xs])
display(Math("U = " + sp.latex(sp.Matrix(np.round(U, 4)))))
print("Problem 1 -- U unitary?", np.allclose(U.conj().T @ U, np.eye(2)))

print("-----")

# Problem 2: completeness sum_i |x_i><x_i| = I
completeness = sum(np.outer(xi, xi.conj()) for xi in xs)
display(Math(r"\sum_i \ket{x_i}\bra{x_i} = " + sp.latex(sp.Matrix(np.round(completeness.real, 4)))))
print("Problem 2 -- equals I?", np.allclose(completeness, np.eye(2)))

print("-----")

# Problem 3: spectral decomposition sum_i lambda_i |x_i><x_i| = sigma_x
spectral = sum(lam * np.outer(xi, xi.conj()) for lam, xi in zip((1, -1), xs))
display(Math(r"\sum_i \lambda_i \ket{x_i}\bra{x_i} = " + sp.latex(sp.Matrix(np.round(spectral.real, 4)))))
print("Problem 3 -- equals sigma_x?", np.allclose(spectral, [[0, 1], [1, 0]]))
```

$$U = \begin{bmatrix} 0.5 + 0.5i & 0.5 - 0.5i \\ 0.5 - 0.5i & 0.5 + 0.5i \end{bmatrix}$$

Problem 1 -- U unitary? True

$$\sum_i |x_i\rangle \langle x_i| = \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}$$

Problem 2 -- equals I? True

$$\sum_i \lambda_i |x_i\rangle \langle x_i| = \begin{bmatrix} 0.0 & 1.0 \\ 1.0 & 0.0 \end{bmatrix}$$

Problem 3 -- equals sigma_x? True

1.3 8. Tensor Products using Colab (Problems 5–7)

```
[3]: # --- Problem 8: verify Problems 5-7 with np.kron ---
f = np.array([0, 1])
e = np.array([1, 0])
g = np.array([1, 1j]) / np.sqrt(2)

# Problem 5: the two orderings differ
fg, gf = np.kron(f, g), np.kron(g, f)
display(Math(r"\ket{f}\otimes\ket{g} = " + sp.latex(sp.Matrix(np.round(fg, 4)).T)))
display(Math(r"\ket{g}\otimes\ket{f} = " + sp.latex(sp.Matrix(np.round(gf, 4)).T)))
print("Problem 5 -- |f>|g> == |g>|f> ?", np.allclose(fg, gf))

print("-----")

# Problem 6: three-fold tensor product |0> (x) |g> (x) |0>
triple = np.kron(np.kron(f, g), f)
print("Problem 6 -- dimension:", triple.shape[0])

print("-----")

# Problem 7: distributivity from both sides
print("Problem 7 -- left :", np.allclose(np.kron(f + e, g), np.kron(f, g) + np.kron(e, g)))
print("Problem 7 -- right :", np.allclose(np.kron(g, f + e), np.kron(g, f) + np.kron(g, e)))
```

$$|f\rangle \otimes |g\rangle = [0 \quad 0 \quad 0.7071 \quad 0.7071i]$$

$$|g\rangle \otimes |f\rangle = [0 \quad 0.7071 \quad 0 \quad 0.7071i]$$

Problem 5 -- |f>|g> == |g>|f> ? False

Problem 6 -- dimension: 8

Problem 7 -- left : True

Problem 7 -- right : True

1.4 11. Tensor Product Calculation using Colab (Problem 10)

```
[4]: # --- Problem 11: verify Problem 10 with np.kron ---
sx = np.array([[0, 1], [1, 0]])
f = np.array([0, 1])
g = np.array([1, 1j]) / np.sqrt(2)

lhs = np.kron(sx @ f, sx @ g) # (sx|f>) (x) (sx|g>)
rhs = np.kron(sx, sx) @ np.kron(f, g) # (sx (x) sx)(|f> (x) |g>)

display(Math(r"(\sigma_x\ket{f})\otimes(\sigma_x\ket{g}) = " + sp.latex(sp.Matrix(np.round(lhs, 4)).T)))
display(Math(r"(\sigma_x\otimes\sigma_x)(\ket{f}\otimes\ket{g}) = " + sp.latex(sp.Matrix(np.
round(rhs, 4)).T)))
print("Problem 11 -- equal?", np.allclose(lhs, rhs))
```

$$(\sigma_x |f\rangle) \otimes (\sigma_x |g\rangle) = [0.7071i \quad 0.7071 \quad 0 \quad 0]$$

$$(\sigma_x \otimes \sigma_x)(|f\rangle \otimes |g\rangle) = [0.7071i \quad 0.7071 \quad 0 \quad 0]$$

Problem 11 -- equal? True
